

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

SEPTEMBER EXAMINATION

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS

TUESDAY, 27 SEPTEMBER 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY

Instruction to Candidates:

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL THREE (3)** questions in **Section A** and choose **ONLY ONE (1)** question in **Section B**.

Should a candidate answer more than **ONE (1)** question in section B, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are not allowed to use calculator.

Answer questions only in the answer booklet provided.

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS

Section A [Total: 75 marks]

This section consists of **THREE (3) COMPULSORY** questions.

- Q1. The following is part of the MegaMax Holding company relational schema which shows the relationship between Employee, Department, Department_Location, Project and Assignment.

EmpID	Fname	Lname	DOB	Address	DateJoined	Sex	BasicSalary	SV_ID	DeptID
S001	Camila	Eleanor	21/05/1990	Ln32, Fenland	01/03/2019	F	\$6000	S008	D004
S002	William	Ethan	14/02/1994	No22, Broadland	02/01/2021	M	\$4500	S007	D003
S003	David	Hudson	23/07/1987	St14, Canterbury	01/06/2017	M	\$8500	S008	D002
S004	Scarlett	Eve	03/11/1989	St21, Cambridge	31/10/2019	F	\$6500	S008	D001
S005	Quinn	Allison	10/04/1992	No8, Birmingham	15/05/2020	F	\$5500	S004	D001
S006	Andrew	Parker	17/03/1990	St11, Boston	15/05/2018	M	\$6000	S003	D002
S007	George	Luis	15/06/1986	Ln44, Cornwall	31/08/2017	M	\$9000	S008	D003
S008	Alex	James	12/03/1980	St43, Boston	01/02/2015	M	\$15000	Null	D004

Table 1.1 Partial contents of the Employee relation

DeptNo	DeptName	ManagerID	Manager_Start Date	Location
D001	Research	S004	01/01/2021	2
D002	Design	S003	31/06/2020	3
D003	Development	S007	01/01/2019	1
D004	Administration	S001	01/03/2019	1

Table 1.2 Partial contents of the Department relation

DeptLocNo	City
1	London
2	Oxford
3	Cardiff

Table 1.3 partial contents of the Department_Location relation

ProjID	ProjName	ProjLocation	DeptID
P10	New Castle	London	D002
P20	New City	Belfast	D003
P32	Product PMX	Glasgow	D001
P40	Mega Prime	London	D003
P43	Product TX	Oxford	D001

Table 1.4 Partial contents of the Project relation

EmpID	ProjNo	Commission	Duration
S003	P10	1.5%	5 months
S006	P10	0.5%	5 months
S005	P32	0.5%	3 months
S004	P43	2.5%	3 months
S007	P20	3.0%	6 months
S004	P32	2.5%	3 months
S002	P40	0.5%	6 months

Table 1.5 Partial contents of the Assignment relation

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q1. (Continued)**

- (a) Write **ONE (1)** Structured Query Language (SQL) statement to list all employees first name, last name and date of joining, who work in the department located at 'London'. (5 marks)
 - (b) Commissions will be given to employees involved in a project by adding a commission rate to the basic salary each month they are involved. Write **ONE (1)** SQL statement to retrieve the employee first and last name for S003, and calculate his total salary over the period he is involved in a project. Name the column as 'Full Name' and 'Total Salary' respectively. (6 marks)
 - (c) Write **ONE (1)** SQL statement to list the first and last names of the managers for each department and the projects handled by their department. (4 marks)
 - (d) Write **ONE (1)** SQL statement to show all the names of employees who joined the company before 2020 and lived in Boston. Arrange the list in the order of employees who joined the company most recently first. (5 marks)
 - (e) Write **ONE (1)** SQL statement to list all first names, last names of employees, number of years worked and the name of their supervisor. Sort the list in ascending order based on the number of years worked. (5 marks)
- [Total : 25 marks]

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS

- Q2. (a) The relational schema shown below is part of the Sports Competition Management database. Each participant can participate in more than one sports competition and each sport can be participated by a maximum of 12 participants.

Participant (participantNo, Name, Country)

Sport (sportID, sportName, sportType, sportCategory, Date)

Result (participantNo, sportID, archivementRecord, Ranking)

- (i) Suggest the appropriate data type for *Date* and *Ranking* attributes from Sport and Result relation respectively. (2 marks)
 - (ii) Identify the primary key and foreign key(s) for each relation. Include the table reference name and the attributes involved for the foreign key. If a relation does not have a foreign key, write NONE. (5 marks)
 - (iii) Identify all parent and child relations from the above relational schema. (3 marks)
 - (iv) Based on the answers in Q2(a)(iii), identify the relationship strength that exists between each relation. (2 marks)
 - (v) Identify **ONE (1)** time-dependent component in Result relation. Justify your answer. (3 marks)
- (b) Table 2.1 shows the list of course registrations by students for the January 2022 trimester at UTAR.

StudentID	FirstName	PhoneNo	RegCourse
S0459	Adele	011-3922819	Statistics, Database Technology
S0199	Carrol	010-1188291	Titas, Programming Concept, Website Development
S0232	Danny	012-5788888	Statistics, Titas, Operating System
S0116	Steve	019-2411421	
S0098	Tina	011-1938471	Project II
S0004	Frankie	011-4632557	Operating System, Mini Project

Table 2.1 Course Registration Records

- (i) State the attribute type for RegCourse. (2 marks)
- (ii) There is a design flaw in Table 2.1. Identify the problem and suggest **ONE (1)** solution for it. (4 marks)

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q2. (Continued)**

- (c) When the relationship in the Figure 2.1 is 1:1, state a possible primary key for ContactInfo relation. Justify your answer. (4 marks)

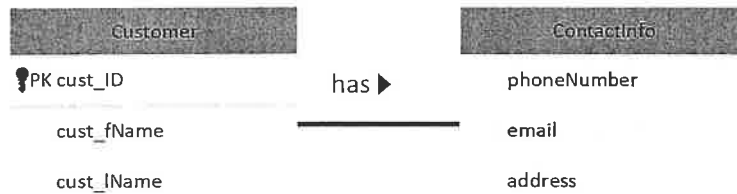


Figure 2.1 Customer/ContactInfo entities

[Total : 25 marks]

- Q3. Figure 3.1 shows an example of book loans information in a secondary school library.

BookLoans

StuID	BookCode	StartDate	EndDate
S001	BC111	12/06/2022	22/06/2022
S002	BC333	01/06/2022	11/06/2022
S002	BC222	09/07/2022	19/07/2022
S003	BC333	15/07/2022	25/007/2022
S001	BC444	21/07/2022	01/08/2022

Book

BookCode	BookName	Quantity
BC111	Sejarah T3	12
BC222	Matematik T5	20
BC333	English T5	10
BC444	Bahasa Melayu T3	15

Student

StudentID	studentName	Classroom
S001	Jasmine	3 Berlian
S002	Michael	5 Intan
S003	Brian	5 Delima

Figure 3.1 Sample Records from Book Loans

- (a) Figure 3.1 shows the tables in a book loans database that have been normalized to the Third Normal Form (3NF) with three entities namely BookLoans, Book and Student.
- Draw the original table in the First Normal Form (1NF). (5 marks)
 - Based on the answer in Q3(a)(i), identify the primary key of the relation. (2 marks)
 - Based on the answer in Q3(a)(i), draw the dependency diagram showing all dependencies in the relation. (5 marks)

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q3. (Continued)**

- (b) Table 3.1 shows the list of suppliers for computer equipment in a production company.

SupplierID	SupplierName	ProductCode	ProductName	Quantity
S001	Digital Enterprise	P010	Printer	10
S001	Digital Enterprise	P004	Laptop	15
S002	FastLine Sdn Bhd	P001	Monitor	25
S003	Global Services	P013	Hard disk	50
S004	Splash Media	P001	Monitor	5
S002	FastLine Sdn Bhd	P005	Mouse	12
S003	Global Services	P010	Printer	8

Table 3.1 List of Computer Equipment Suppliers

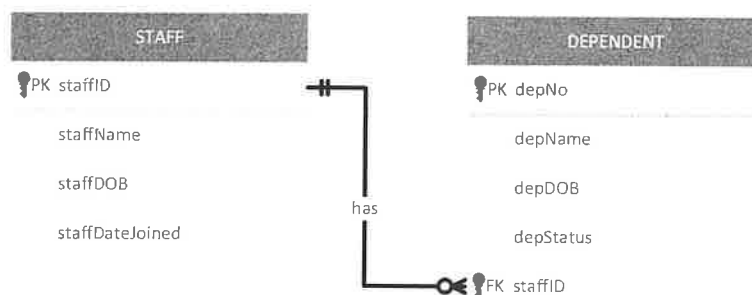
- (i) Based on Table 3.1, identify **THREE (3)** update anomalies in the records. Provide an example for each anomaly. (6 marks)
- (ii) Normalize the relation to Third Normal Form (3NF). Label the primary key and foreign key in each relation. (7 marks)
- [Total : 25 marks]

Section B [Total: 25 marks]

This section consists of **TWO (2)** questions. Answer **ONE (1)** question only.

- Q4. (a) There are several strategies to consider when there is a child occurrence referencing to the parent occurrence that is trying to be deleted. Appropriate action depends on the tables involved. Consider the following ERD (Entity Relationship Diagram), identify the parent and the child, then indicate the appropriate action if the parent occurrence is deleted. Justify your answer.

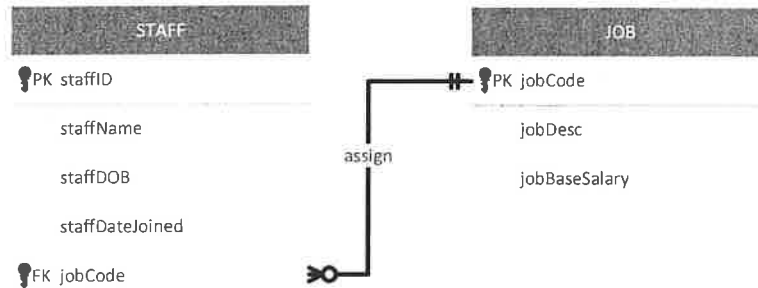
- (i)



(5 marks)

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS**Q4(a). (Continued)**

(ii)



(5 marks)

- (b) MediCare Clinic is a new clinic in Kampar town that started operations in early 2022. To streamline the operation of the clinic, the management has set some regulations. For appointment arrangements, a patient is allowed to make several appointments with one or more doctors at the clinic, and one doctor can accept appointments with many patients. However, for each appointment, only one doctor and one patient are allowed. In the event of an emergency, an appointment is not necessary. Nevertheless, it still needs to be recorded and labeled as unscheduled. Once the appointment session is confirmed, the consultation will take place in the presence of doctor and patient who have been specified in the appointment. The doctor will make a diagnosis and when appropriate, treatment will be suggested and carried out. Each consultation with the patient will be billed. Each bill will be issued by one doctor, and each doctor can issue a bill for many patients. Each bill must be paid within the stipulated period. However, bills can be paid in installments, and each payment can cover more than one bill.

Draw an ERD from the above scenario that includes all the entities, attributes of each entity, relationships, primary key, foreign key(s) and the multiplicity of the relationship, i.e., cardinality and optionality. For each entity, list at least **THREE (3)** relevant attributes except for those which have been assigned. Use **CROW'S FOOT notation** or **UML notation** to illustrate your ERD.

(15 marks)

[Total : 25 marks]

UCCD1203 DATABASE DEVELOPMENT AND APPLICATIONS

- Q5. (a) Figure 5.1 shows an invoice received by a customer after purchasing items at a store selling computer products and accessories. Each invoice will be generated for each purchase by a customer and issued by an employee. There are three payment methods allowed, namely cash, debit/credit card or e-wallet.

Draw an Entity Relationship Diagram (ERD) from the above scenario that includes all the entities, attributes of each entity, relationships, primary key, foreign key(s) and the multiplicity of the relationship, i.e., cardinality and optionality. For each entity, list at least **THREE (3)** relevant attributes except for those which have been assigned. Use **CROW'S FOOT notation** or **UML notation** to illustrate your ERD. (15 marks)

SPEnterprise

No12, Jalan Semarak 2,
Taman Perniagaan Damai,
31200 Ipoh, Perak.
Tel: 05-3938182 Fax: 05-3938182

Customer Invoice

Bill To: Caroline Eve
No42, Palace Condominium,
31200 Ipoh, Perak.

Tel: 012-5932817

Invoice No : SP002215
Date : 26/05/2022
Payment : Cash
Salesman : 002
Issue By : Edward

NO.	DESCRIPTION	QTY	PRICE PER UNIT	AMOUNT (RM)
1.	Laptop X360	1	2500	2500
2.	Pendrive 64GB	2	60	120
3.	Mouse	1	45	45

TOTAL: RM2665

Figure 5.1 Customer Invoice

- (b) There are two types of connection traps that often occur due to misinterpretation of the meaning of a particular relationship between entities. Draw a simple ERD to illustrate the following connection traps.

(i) Fan trap. (5 marks)

(ii) Chasm trap. (5 marks)

[Total : 25 marks]